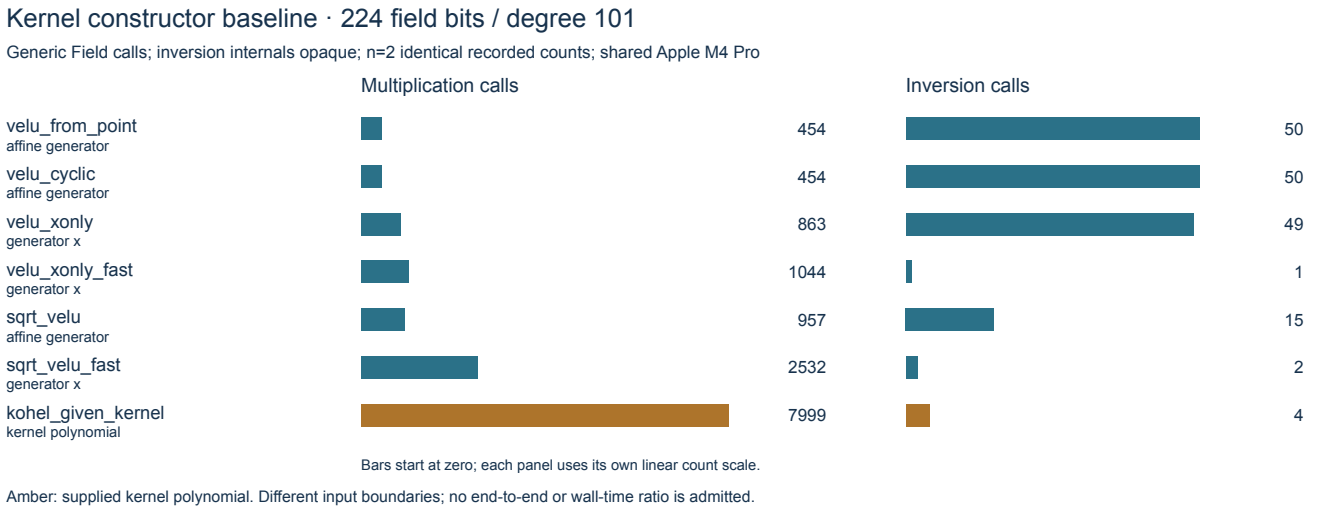
Seven kernel constructor baselines

2026-10-15 · Two repeated native 84-case runs

Seven methods at 192/224/256/511 field bits, degrees 7/31/101. All 84 paired output fingerprints and all four field-call counters repeat exactly. Synthetic supersingular fixtures; these are not the named NIST curve models.

Measured calls at 224 bits / degree 101

Method	Supplied input	Mul calls	Inv calls
velu_from_point	affine generator	454	50
velu_cyclic	affine generator	454	50
velu_xonly	generator x	863	49
velu_xonly_fast	generator x	1044	1
sqrt_velu	affine generator	957	15
sqrt_velu_fast	generator x	2532	2
kohel_given_kernel	kernel polynomial	7999	4





Fixture/reference construction and verification excluded. Native representation allocation included. No wall-time admission.

Generic Field API calls, inversion internals opaque; n=2 identical recorded counts. Different supplied-input boundaries remain separate. Reference/fixture generation, fingerprints and verification are excluded; owned representation allocation is included. These baseline counts establish no kernel speedup.

Degree, codomain and full point transports at 2ell+1 distinct non-kernel abscissae agree with both Kohel and point-enumerating Vélu. Counts are saved before verification. These fixture fingerprints are distinct from the BMSS coefficient comparison.

Shared Apple M4 Pro, 14 logical CPUs, 48 GiB, macOS ARM64. Zero timing samples requested or qualified. Other physical CPU baselines and optimization rounds remain open.

Source: 5521605f86ae444a9f4a433ca9b2f80d541e8470 Binary SHA256: e22a305acbd0ab58be40a7fe387867f10598255b3cf5ec03532d98fbb1b50c6e Comparison SHA256: 80dc2ebcf465fade24a71e55c47f366feed3ab200f1136196572b7ee8034c496

Read REPORT.md, COUNTS.csv and KERNEL_BASELINE_PROTOCOL.md with the frozen raw records and source/binary archive. Both complete matrices, including all other widths and degrees, remain in the raw comparison.

Kohel's result contains all rational-map coefficients. The other constructor results store compact evaluation representations. Their build counts include different output materialization costs.